

CDIM — Cascading Dependency Impact Module

Parent Standard: Operational Dependency & Coordination Standard (ODCS)

Category: Governance & Enforcement

Subcategory: Cascading Dependency Impact

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Compatibility: OOF Methodology OS · Operational Dependency & Coordination Standard (ODCS) · Operational Reality Standard (ORS) · Runtime Integrity Standard (RIS) · Operational Evidence & Auditability Standard (OEAS) · Semantic Integrity Standard (SEIS) · Authority & Accountability Layer Standard (AALS) · INTEGROS® — Integrity Standard · Multi-Layer Truth Validation Framework (MTVF) · Ethical Virtual Integrity Protocol (EVIP)

Authority: OOF

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Canonical Language: English (UCL)

Canonical Definition

Cascading Dependency Impact Module (CDIM) defines the structural conditions under which cascading operational dependency effects remain materially identifiable, governable, traceable, and operationally containable across interconnected runtime environments, orchestration systems, delegated execution chains, and consequence-bearing operational infrastructures.

A system satisfies CDIM only if:

- cascading dependency effects remain materially identifiable
- dependency-impact propagation remains governable
- runtime cascade continuity remains traceable
- operational fragmentation does not silently propagate across dependency chains
- consequence-bearing cascading effects remain operationally containable

A system that preserves local operational continuity while cascading dependency instability silently propagates across interconnected environments does not satisfy CDIM.

Module Operational Role

CDIM defines the cascading dependency governance layer of operational coordination governance by preserving governable containment of cascading operational fragmentation across dependency-bound runtime environments.

Module Operational Space

CDIM governs the cascading dependency impact space of ODCS by preserving cascading operational traceability, dependency-impact containment continuity, runtime propagation visibility, orchestration cascade governance, and consequence-bearing dependency stabilization across distributed operational systems.

Module Function

The module applies wherever systems must preserve:

- cascading dependency traceability
- runtime propagation visibility
- orchestration cascade stability
- distributed dependency containment
- consequence-bearing dependency governance
- operational fragmentation containment continuity

Its function is to ensure that cascading operational dependency effects remain materially governable strongly enough to prevent uncontrolled operational fragmentation across interconnected systems.

Minimum Implementation Framework

1. Define the Cascading Dependency Object

The organization must define which dependency relationships or operational propagation chains require cascade- governance preservation.

This may include:

- orchestration dependency chains
 - runtime propagation paths
 - delegated dependency cascades
 - operational synchronization chains
 - distributed execution propagation
 - consequence-bearing dependency relationships
 - interconnected operational infrastructures
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2. Define Cascading Dependency Conditions

The system must define the conditions under which cascading dependency effects remain materially governable and operationally containable.

This includes:

- propagation continuity visibility
 - dependency-impact traceability
 - orchestration containment continuity
 - distributed stabilization continuity
 - runtime cascade governance continuity
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3. Define Cascade Fragmentation Detection Logic

The system must define how materially unstable cascading dependency conditions or propagation fragmentation is identified.

This may include:

- uncontrolled dependency propagation
- hidden orchestration instability
- runtime cascade divergence
- distributed fragmentation propagation
- consequence-bearing dependency escalation

4. Define Operational Response or Governance Logic

The system must define governance logic for materially unstable cascading dependency conditions.

Governance response may include:

- cascade escalation
- runtime isolation
- orchestration narrowing
- dependency containment activation
- propagation stabilization enforcement
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of cascading dependency continuity and cascade-fragmentation states. A system must not remain coordination-valid if cascading dependency instability materially propagates across interconnected environments while systems continue assuming stable operational coordination remains preserved.

Use Case 1 — AI Orchestration Cascade

Propagation

Scenario

A distributed AI orchestration environment experiences cascading runtime instability across multiple interconnected agents, orchestration systems, and dependency-bound operational chains.

Application

CDIM preserves governable containment and traceable propagation continuity across cascading runtime dependency environments.

Result

The environment gains stronger cascade containment stability and reduced uncontrolled dependency propagation across AI operational systems.

Use Case 2 — Enterprise Dependency

Infrastructure

Scenario

An enterprise operational infrastructure depends on interconnected systems where runtime instability in one environment may propagate across dependent operational chains.

Application

CDIM preserves traceable containment continuity across cascading dependency propagation environments.

Result

The organization gains stronger operational resilience and reduced cascading fragmentation risk across distributed operational infrastructures.

Canonical Closing Statement

If cascading operational dependency effects cannot remain materially governable across interconnected environments, systems may preserve local operational continuity while operational fragmentation silently propagates underneath.

Related Documents

→ [Operational Dependency & Coordination Standard - \(ODCS\)](#).

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